DrainGuard

AUTONOMOUS INTELLIGENT MODULAR REAL-TIME

Autonomous Robotic System for Stormwater Drain Inspection and Blockage Detection

DrainGuard is an intelligent mobile robotic system designed to inspect stormwater drains, identify blockages and infrastructure issues, monitor environmental conditions, and provide real-time visual and sensor data — without requiring humans to enter hazardous drainage environments.

Team CyberTron

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GitHub: https://github.com/LuckyKashyap001/DrainGuard---Autonomous-Robotic-System-for-Stormwater-Drain-Inspection-and-Blockage-Detection.git

Future Scope Highlights

- Fully autonomous AI-based path planning and navigation
- Advanced blockage classification and damage detection models
- Autonomous blockage-removal robotic manipulator
- Digital drain mapping and predictive maintenance analytics
- Multi-robot systems and smart-city cloud integration

Challenges in Stormwater Drain Inspection

Stormwater drainage systems are critical for preventing urban flooding, but regular inspection is difficult. Many drains remain unmonitored until blockages, structural damage, or water accumulation cause serious problems.

1. Hazardous Manual Inspection

Workers face toxic gases, contaminated water, low oxygen levels, darkness, and unsafe confined spaces — creating significant occupational health and safety risks with every entry.

2. Delayed Blockage Detection

Plastic, leaves, mud, sediment, and solid waste gradually block water flow. Problems are often discovered only after flooding occurs, leading to reactive rather than proactive maintenance.

3. Difficult Accessibility

Many drains and pipelines are narrow, underground, poorly illuminated, and difficult for maintenance personnel to access safely or efficiently — limiting inspection frequency.

4. Lack of Real-Time Data

Traditional inspection provides limited continuous information about blockage conditions, environmental parameters, and infrastructure health. Data collection is manual and time-consuming.

5. High Operational Effort

Manual inspection requires trained personnel, safety equipment, repeated field visits, and significant time investment — increasing operational costs and reducing inspection frequency.

The Core Problem: A safer, faster, and data-driven inspection system is required to detect problems before they become major infrastructure failures.

DrainGuard: A Smart Robotic Inspection Platform

DrainGuard transforms drain inspection from a manual, hazardousprocess into an intelligent, data-driven operation—entering drainage infrastructure to perform visual inspection, environmental monitoring, blockage detection, and real-time data transmission.

Autonomous / Remote Mobility

Four-wheel design provides stability on uneven, wet, and debris-filled surfaces with remote or autonomous navigation.

Real-Time Camera Inspection

Dual cameras enable operators to inspect dark, narrow, inaccessible areas without physical entry via live video streaming.

AI-Based Detection

Computer vision analyzes camera feeds to identify blockages, debris, leakage, and damage with confidence scoring and severity estimation.

Environmental Monitoring

Sensors monitor temperature, humidity, gas conditions, tilt, battery status, and operational parameters for safe operation.

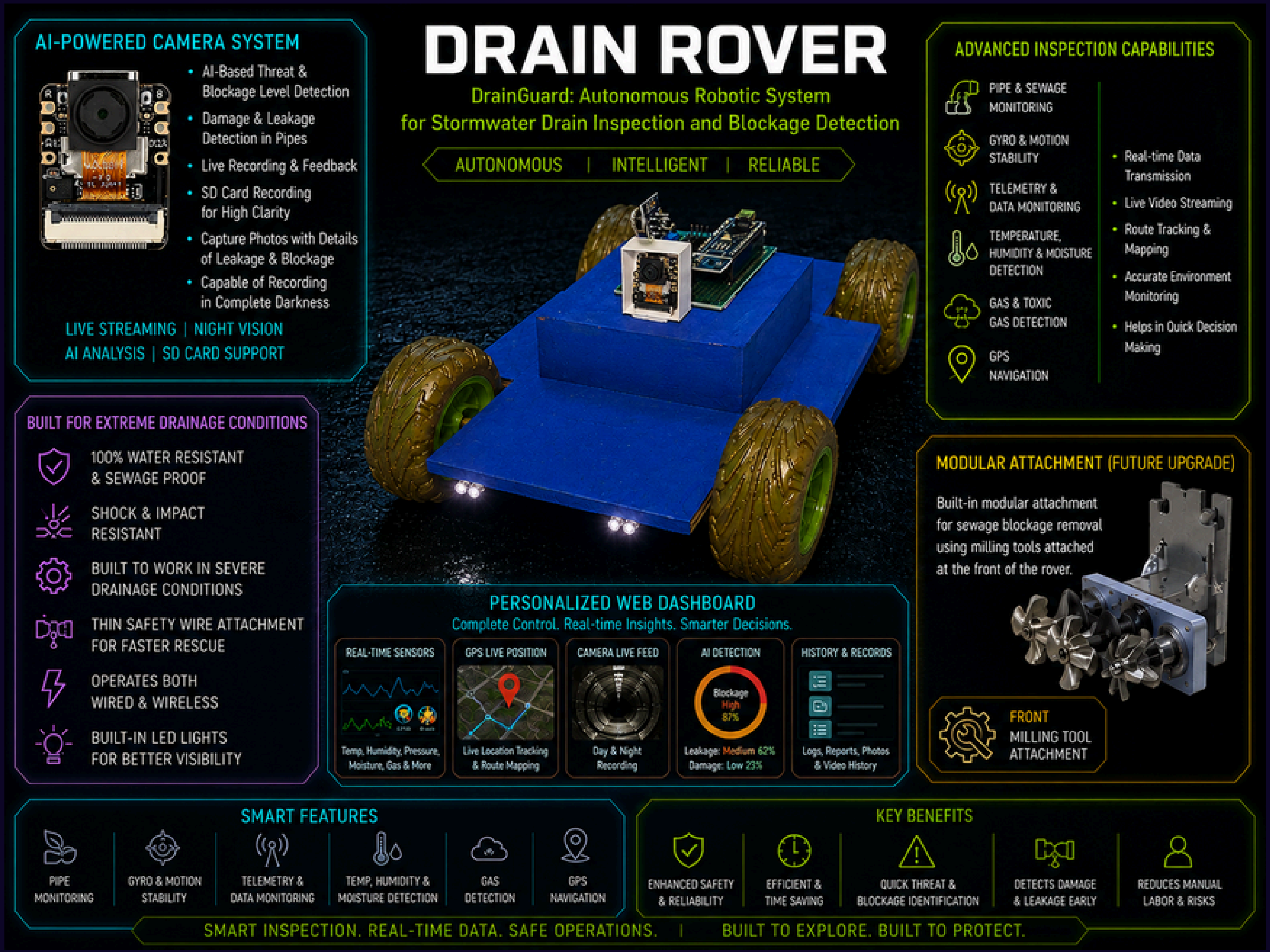
Web-Based Monitoring

Centralized dashboard with live video, sensor readings, AI alerts, and mission information for remote monitoring and control.

Modular Architecture

Future attachments extend the rover from inspection into blockage-removal and maintenance, supporting long-term scalability.

Prototype Overview



Note : This is just a prototype developed for proof-of-concept and system validation, focusing on verifying sensor readings, testing component integration and core functionalities, evaluating communication and control performance, assessing dimensional adaptability, and identifying potential improvements before the final system implementation.

Robot Design & Hardware Architecture

DrainGuard features arobust mobile platform engineered for reliable operation in challenging drainage environments — prioritizing stability, durability, and accessibility while maintaining a compact form factor suitable for narrow inspection spaces.

Mechanical Design Features

- **Four-Wheel Mobile Platform:** Improved stability and traction on uneven surfaces
- **Rugged Wheels:** Designed for wet, muddy, and debris-filled drainage environments
- **Compact Body:** Suitable for narrow inspection spaces and confined areas
- **Low Center of Gravity:** Improves movement stability and reduces tipping risk
- **Protected Electronics Enclosure:** Waterproof and sealed for harsh field conditions
- **LED Illumination System:** Enables operation in dark drainage environments
- **Modular Front Section:** Supports future maintenance tools and blockage-removal attachments

Main Hardware Components

Component	Function
Main Controller (ESP32)	Processes sensor data and controls rover operations
Camera Module	Provides live inspection video and image capture
AI Processing Unit	Detects blockages, debris, leakage, and anomalies
Motor Driver & Motors	Controls direction and speed of the rover
IMU / Gyroscope	Measures orientation, tilt, and movement
Gas Sensors	Detects potentially hazardous atmospheric conditions
Temp & Humidity Sensors	Monitors surrounding environmental conditions
GPS Module	Provides outdoor position and route information
Wireless Module (WiFi/4G)	Sends data to remote dashboard in real-time
Battery System	Powers rover during inspection missions
LED System	Improves visibility in dark drainage areas
Modular Attachment Port	Supports future blockage removal and maintenance tools

AI-Powered Camera System

DrainGuard employs a dual-camera system combining real-time navigation feedback with intelligent blockage detection. The AI pipeline processes live video to identify drainage problems and provide actionable alerts to operators.

Front Camera — Navigation & General Inspection

- Real-time navigation feedback for operator control
- Obstacle observation and avoidance assistance
- Pipe and drain wall inspection with detailed visibility
- Detection of visible damage, cracks, and structural issues
- Leakage observation and water flow assessment
- Live video streaming with low-latency transmission
- Image and video capture for documentation
- Operation in low-light conditions using LED illumination

Inspection Camera — AI-Based Blockage Detection

- Detect debris and waste accumulation with high accuracy
- Identify potential blockages before they become critical
- Recognize mud and sediment buildup patterns
- Detect plastic and solid waste materials
- Estimate blockage severity and urgency level
- Identify visible leakage or structural anomalies
- Generate AI confidence scores for each detection
- Classify blockage types for targeted maintenance

Live Feed → **Frame Capture** → **Preprocessing** → **AI Detection**

🔴 AI DETECTION ALERT

Object: Drain Blockage
Confidence: 87%
Severity: HIGH
Action: Inspection / Cleaning Required

Capabilities

🔴 Live Streaming
🌙 Night Inspection
🤖 AI Detection
📷 Photo Capture
🎥 Video Recording

Sensor System & Real-Time Telemetry

DrainGuard combines visual intelligence with comprehensive environmental and operational sensing. Real-time telemetry provides operators with complete awareness of both the drain environment and the rover's operational status during inspection missions.

🌡

Environmental Monitoring

- **Temperature:** Monitors drain environment temperature for safety assessment
- **Humidity:** Tracks moisture levels to identify water accumulation
- **Condition Analysis:** Assesses overall drain environment health and stability

⚠

Safety Monitoring

- **Gas Detection:** Detects methane, hydrogen sulfide, carbon dioxide
- **Environmental Alerts:** Triggers warnings when readings exceed safe thresholds
- **Operator Protection:** Prevents unsafe operation in hazardous conditions

Motion & Stability

- **Gyroscope / IMU:** Monitor tilt and orientation during movement
- **Stability Assessment:** Identifies unstable movement or difficult terrain
- **Navigation Assistance:** Helps operators understand rover position

Navigation & Position

- **GPS Tracking:** Records outdoor deployment location and route
- **Route Monitoring:** Supports mission documentation and coverage analysis
- **Location History:** Enables future autonomous navigation and mapping

Live Telemetry Panel

Parameter	Reading
🌡️ Temperature	29°C
💧 Humidity	84%
📐 Tilt	4.2°
🔋 Battery	78%
Gas Status	NORMAL
📍 GPS	CONNECTED
📷 Camera	ONLINE
🤖 AI Status	BLOCKAGE DETECTED

Sensor data is continuously transmitted to the dashboard, allowing the operator to understand both the physical drain environment and the operational condition of the robot during a mission.

Web Dashboard Overview

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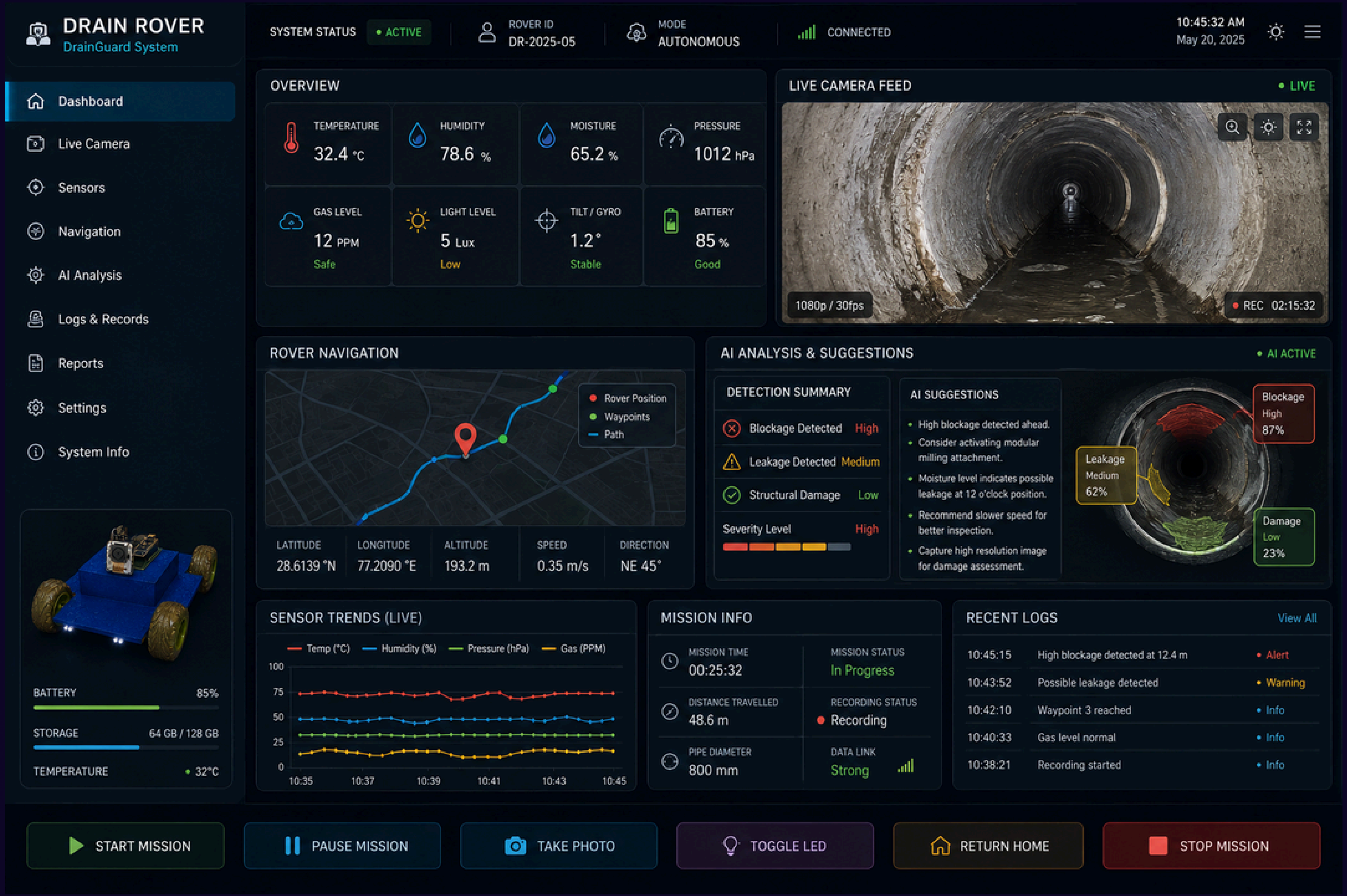

Note : The web dashboard preview is presented for demonstration purposes to showcase the proposed interface, live sensor monitoring, AI insights, telemetry, and system control features.

Working Principle & Operation

DrainGuard performs structured inspection missions through a systematic 8-step process — converting drain inspection into a continuous digital workflow where problems can be identified, recorded, and acted upon more efficiently than traditional manual methods.

1. **Step 1 — Deploy**
   Rover placed at inspection location; operator verifies physical condition and positioning at drain entrance.
2. **Step 2 — System Check**
   Battery, cameras, sensors, motors, and communication systems verified. All subsystems must report operational status.
3. **Step 3 — Connect Dashboard**
   Operator connects to web dashboard to monitor live video, telemetry, camera feeds, and AI detection status.
4. **Step 4 — Navigate**
   DrainGuard moves through the inspection area via remote control or assisted navigation while monitoring obstacles.
5. **Step 5 — Collect Data**
   Cameras and sensors collect live information. Front camera provides navigation; inspection camera focuses on blockage detection.
6. **Step 6 — AI Analysis**
   AI processes camera frames to identify debris, blockages, leakage, and infrastructure problems with confidence scoring.
7. **Step 7 — Alert & Log**
   Important events displayed on dashboard with visual and audio alerts. Images, videos, sensor logs, and AI detections stored.
8. **Step 8 — Decide & Act**
   Maintenance team decides whether cleaning, blockage removal, repair, or additional inspection is required based on digital records.

1 DEPLOY

2 CONNECT

3 NAVIGATE

4 INSPECT

5 ANALYZE

6 ACT

Web Dashboard & Remote Control

The DrainGuard web dashboard provides a centralized interface for mission monitoring and rover control. Operators can view live video, monitor sensor data, respond to AI alerts, and control rover movement from a single professional interface.

📹 Live Camera Feed

- Front camera for navigation and general inspection
- Inspection camera for blockage detection focus
- Low-latency video streaming and recording status
- Screenshot, photo capture, and full-screen mode

🤖 AI Detection Panel

- Detected object type and classific ation
- Confidence level percentage per detection
- Severity rating: Low / Medium / High / Critical
- Detection history, timeline, and recommended actions

📊 Sensor Telemetry Display

- Temperature, humidity, gas status, and hazard alerts
- Tilt, orientation, battery level, and power consumption
- System health status — real-time updates every 500ms

📍 GPS & Mapping Module

- Current rover position and route tracking
- Mission location and area coverage visualization
- Waypoint marking and navigation history

🕹️ Rover Control Interface

- Forward, Reverse, Left, Right directional controls
- Stop and Emergency Stop buttons
- LED brightness control and camera switching
- Video recording start/stop and speed adjustment slider

🗂️ History & Records Module

- Captured images and video files archive
- Sensor data logs and telemetry records
- AI detection events and alert history
- Inspection reports, mission timestamps, and export functionality

Key Benefit: The dashboard gives the operator a single interface for controlling the rover, observing the drain environment, monitoring sensor data, and responding to AI-generated alerts in real-time.

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Innovation, Benefits & Future Upgrades

DrainGuard representsa significant advancement in drainage infrastructure management—combining robotics, AI, and IoT to transform inspection from a manual, hazardous process into an intelligent, data-driven operation. The modular architecture enables continuous evolution toward fully autonomous smart-city infrastructure management.

Why DrainGuard is Valuable

Improves Human Safety

Eliminates exposure to toxic gases, contaminated water, and confined space hazards for maintenance personnel.

Enables Early Detection

AI inspection identifies developing blockages before they lead to severe flooding and emergency response costs.

Provides Real-Time Information

Live video and sensor readings enable faster decision-making and more effective maintenance planning.

Creates Digital Records

Images, videos, telemetry, and AI events documented for trend analysis and predictive maintenance strategies.

Modular and Scalable

Single-robot inspection can evolve into multi-robot systems for large drainage networks across entire cities.

Future Upgrade Roadmap

01

Fully AutonomousNavigation

AI-based path planning, obstacle avoidance, and autonomous exploration without operator control.

02

SLAM & Digital Mapping

Create detailed 3D maps of underground drainage infrastructure for future reference and planning.

03

Advanced AI Models

Classify blockage types, leakage patterns, cracks, corrosion, and structural damage with higher accuracy.

04

Autonomous Blockage Removal

Front-mounted milling, cutting, or debris-clearing mechanism for automated maintenance operations.

05

Cloud & Smart City Integration

Centralized monitoring of multiple drainage locations across entire cities with predictive analytics.

06

Multi-Robot System

Multiple DrainGuard units inspecting large drainage networks simultaneously with autonomous charging.

1

CURRENT PROTOTYPE

2

SMART INSPECTION

3

AUTONOMOUS DRAIN ROBOT

4

CONNECTED SMART CITY

DrainGuard: Smarter Inspection for Safer Drainage Infrastructure

"Inspect Earlier. Detect Smarter. Protect Better."

Project Summary

🤖 Mobile Robotics

Four-wheel platform for accessing difficult drainage environments safely

🧠 AI-Powered Vision

Intelligent blockage detection and anomaly classification with confidence scoring

📷 Dual Camera

Navigation and detailed monitoring capabilities in dark, confined spaces

🌡️ Environmental Sensors

Real-time condition monitoring, gas detection, IMU, and GPS tracking

📡 Real-Time Telemetry

Continuous mission awareness and data transmission to web dashboard

🔧 Modular Architecture

Future upgrades including blockage-removal and autonomous capabilities

Innovation Stack

- Robotics
- AI & Computer Vision
- IoT & Telemetry
- Web Technology

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THANK YOU